ASSIGNMENT

Name	Gorantla Venkata Hima Sandeep
Registration No.	99230041051
Year	IV
Department	CSE(AIML)

Designing an AI-Driven Multi-Step Prompt Workflow

Objective

To design a multi-step prompt workflow for an AI-driven healthcare appointment assistant. The workflow uses AI to understand a patient's message, identify important information, assess urgency, and generate a suitable appointment recommendation.

Real-World Example

A hospital uses an AI-driven appointment assistant to help patients book appointments through a chat interface. The assistant receives the following message:

"I have been experiencing a severe headache since yesterday. I also feel dizzy and have difficulty concentrating. I would like to meet a doctor as soon as possible. Please help me book an appointment."

The AI system must understand the message, extract the symptoms and appointment requirement, assess the urgency, and generate a safe and helpful response. The AI must not provide a medical diagnosis. It should recommend urgent professional help when necessary.

AI Used in the Workflow

The workflow uses Artificial Intelligence and Natural Language Processing to understand the patient's message. A language model extracts relevant details, classifies the urgency using predefined rules, and generates a natural-language response. The final appointment decision should follow hospital policy and, when required, be reviewed by a healthcare professional.

Workflow Overview

Step 1	AI extracts patient information
Step 2	AI assesses appointment urgency

Step 3	AI generates a safe appointment response

Step 1: Extract Patient Information Using AI

Purpose

The first AI prompt converts the patient's natural-language message into structured information. This makes the next step easier and reduces the chance of missing important details.

Prompt

```
You are an AI healthcare appointment information-extraction assistant.

Read the patient's message and extract the following information:

1. Main complaint
2. Symptoms mentioned
3. Duration of symptoms
4. Patient's request
5. Preferred urgency
6. Any emergency warning signs explicitly mentioned

Do not diagnose the patient. Return the information in a clear
key-value format.

Patient message:
"I have been experiencing a severe headache since yesterday.
I also feel dizzy and have difficulty concentrating. I would like
to meet a doctor as soon as possible. Please help me book an appointment."
```

Expected Output

```
Main complaint: Severe headache
Symptoms mentioned: Dizziness and difficulty concentrating
Duration of symptoms: Since yesterday
Patient's request: Book a doctor appointment
Preferred urgency: As soon as possible
Emergency warning signs explicitly mentioned: None stated
```

Step 2: Assess Appointment Urgency Using AI

Purpose

The second AI prompt evaluates the extracted information according to predefined hospital rules. It does not diagnose the patient. It only recommends the appropriate level of attention.

Prompt

```
You are an AI healthcare appointment-priority assistant.

Classify the appointment request as Routine, Priority, or Emergency
based only on the information provided.

Rules:
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- Routine: General consultation or non-urgent symptoms.
- Priority: Symptoms that need medical evaluation soon or are affecting
normal activities.
- Emergency: Severe or rapidly worsening symptoms, loss of consciousness,
difficulty breathing, chest pain, signs of stroke, or other emergency
warning signs. If emergency signs are present, advise the patient to
contact local emergency services immediately.

Do not provide a diagnosis. Explain the classification briefly.

Extracted information:
Main complaint: Severe headache
Symptoms mentioned: Dizziness and difficulty concentrating
Duration of symptoms: Since yesterday
Patient’s request: Book a doctor appointment
Preferred urgency: As soon as possible
Emergency warning signs explicitly mentioned: None stated
```

Expected Output

```
Urgency: Priority

Reason: The patient reports a severe headache along with dizziness
and difficulty concentrating. These symptoms should be evaluated
by a healthcare professional soon. The AI should not provide a
diagnosis or assume that the condition is an emergency without
additional information.

Recommended handling: Offer the earliest available appointment
and advise the patient to seek emergency help if severe or
life-threatening warning signs develop.
```

Step 3: Generate the Final AI Response

Purpose

The third prompt uses the extracted information and urgency assessment to generate a polite, safe, and useful response for the patient.

Prompt

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You are an AI-driven hospital appointment assistant.

Generate a safe and professional response using the information below.

The response must:
1. Acknowledge the patient’s symptoms.
2. Avoid giving a diagnosis or prescribing treatment.
3. Recommend the earliest available doctor appointment.
4. Ask the patient to provide suitable appointment details if needed.
5. Include a clear safety instruction to seek emergency help if
severe or life-threatening symptoms occur.
6. Use the following format:

Patient Concern:
Urgency:
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AI Response:
Recommended Action:

Extracted information:
Main complaint: Severe headache
Symptoms mentioned: Dizziness and difficulty concentrating
Duration of symptoms: Since yesterday
Patient’s request: Book a doctor appointment
Preferred urgency: As soon as possible
Emergency warning signs explicitly mentioned: None stated

Urgency:
Priority
```

Expected Output

```
Patient Concern: Severe headache with dizziness and difficulty concentrating

Urgency: Priority

AI Response:
I’m sorry that you are experiencing these symptoms. Based on the
information provided, your request should be handled as a priority.
I can help you find the earliest available appointment with a
doctor. Please provide your preferred date, time, and hospital
location, if applicable.

This assistant cannot provide a diagnosis. If your symptoms become
severe, suddenly worsen, or you experience an emergency warning
sign such as difficulty breathing, loss of consciousness, chest
pain, or sudden weakness, contact local emergency services or
visit the nearest emergency department immediately.

Recommended Action:
Offer the earliest available appointment and escalate the request
to a healthcare professional according to hospital policy.
```

Complete Prompt Workflow

Patient Message	Input received through the hospital chat system
Step 1	AI extracts symptoms, duration, and appointment request
Step 2	AI classifies urgency using hospital-defined rules
Step 3	AI generates a safe and structured response
Final Action	Appointment booking or professional escalation

Advantages of the AI-Driven Workflow

• AI can understand patient messages written in natural language.

• The workflow separates information extraction, reasoning, and response generation.

• Structured outputs make the system easier to evaluate and integrate with hospital software.

• Predefined rules help maintain consistency in urgency classification.

• The assistant can reduce the workload of appointment-support staff.

• The workflow can be reused for appointment booking, follow-up requests, prescription queries, and general hospital information.

Safety and Limitations

• The AI assistant must not diagnose diseases.

• The AI must not replace a qualified healthcare professional.

• Emergency symptoms must be escalated according to hospital policy.

• Patient information must be protected and handled securely.

• The final appointment or emergency decision should be reviewed by authorized staff when necessary.

Conclusion

This AI-driven multi-step prompt workflow demonstrates how Artificial Intelligence can support hospital appointment services. The first step extracts useful information from the patient’s message, the second step assesses urgency using predefined rules, and the third step generates a safe response. This approach makes the AI system more organized, reliable, and useful while ensuring that important medical decisions remain under professional supervision.